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AWS-DEVOPS



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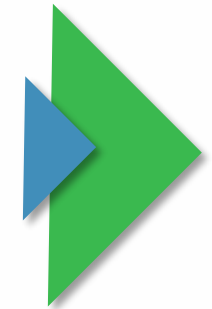


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Command Line Basics - Day 2



Learning Objectives



- Manage files and directories safely with `rmdir`, `rm`, `cp`, and `mv`.
- Create and inspect simple file content with `echo` and `cat`.
- Test reachability and connect to Linux servers with `ping` and `SSH`.
- Download artifacts with `wget` and `curl`.
- Archive/extract files with `tar` and `unzip`.
- Find command documentation with `--help`, `man`, `info`, `type`, and `which`.

Day 2 focus: practical commands you will repeatedly use on EC2 and in DevOps workflows.

Removing Files and Directories



rm

Removes an empty directory.

```
mkdir empty-dir  
rm empty-dir
```

rm

Removes files. Linux CLI deletion normally bypasses a recycle bin.

```
touch old.log  
rm old.log
```

**Verify the path before deleting.
Recovery may be difficult or
impossible.**

Using rm Safely



```
rm -i file.txt           # ask before deleting
rm -r old-dir/           # remove directory
recursively
rm -rf old-dir/          # recursive + force
```

- `-i` is useful while learning because it asks for confirmation.
- `-r` recursively removes a directory and its contents.
- `-f` suppresses many prompts/errors; combine with `-r` only when you are certain.

DevOps habit: run `pwd` and `ls` before destructive commands, especially when using `sudo`.

Copying Files and Directories



```
cp source.txt backup.txt  
cp source.txt /tmp/  
cp -r config/ config-backup/
```

- `cp` copies a file to a new name or location.
- `cp -r` recursively copies directories.
- The original remains in place after a copy.

Common DevOps use: copy a configuration file before editing it.

```
sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.bak
```

Moving and Renaming



```
mv app.log logs/  
mv old-name.txt new-name.txt
```

- `mv` moves a file or directory to another location.
- The same command also renames files and directories.
- Unlike `cp`, the original path no longer remains after a move.

Tip: use `mv -i` when you want a prompt before overwriting an existing destination.

```
mv -i app.conf config/app.conf
```

echo and Output Redirection



```
echo "hello DevOps"  
echo "PORT=8080" > app.env  
echo "DEBUG=false" >> app.env
```

- ``echo`` prints text to standard output.
- ``>`` is shell redirection: create/overwrite the target file.
- ``>>`` appends output to the end of the target file.

Important: ``>`` is performed by the shell; it is not an option of echo.

```
cat app.env
```

Using cat



```
cat app.env  
cat file1.txt file2.txt  
cat file1.txt file2.txt > combined.txt
```

- `cat` prints file contents to standard output.
- It can concatenate multiple files.
- For large files, tools such as `less` are usually easier to navigate (covered later).

Use cat for quick inspection of small configuration and text files.

Mini-Lab: Manage an Application Directory



```
mkdir -p app/{config,logs,backup}  
touch app/config/app.conf app/logs/app.log  
cp app/config/app.conf app/backup/app.conf.bak  
mv app/logs/app.log app/logs/application.log  
ls -R app
```

- Create a simple project structure.
- Back up a configuration file.
- Rename a log file.
- Verify the final structure.

Done condition: app/config, app/logs and app/backup exist with the expected files.

File Management Quick Reference



Command	Purpose	Example
rmdir	remove empty directory	rmdir temp
rm	remove file	rm old.log
rm -r	remove directory recursively	rm -r old/
cp	copy file	cp a.txt b.txt
cp -r	copy directory	cp -r app/ backup/
mv	move or rename	mv a.txt archive/
echo	print text	echo "hello"
cat	show/concatenate text	cat app.conf

Task: Build a Small DevOps Workspace



Create this structure:

```
devops-workspace/  
├── config/  
│   └── app.conf  
├── logs/  
│   └── app.log  
└── backup/
```

- Create all directories.
- Write APP_ENV=dev into app.conf.
- Copy app.conf into backup as app.conf.bak.
- Rename app.log to application.log.
- Display the final structure with ls -R.

Try it without copying commands from the previous slides.



Networking & Remote Access



ping: Test IP Reachability



ping sends ICMP Echo requests and reports whether replies are received, along with round-trip time.

```
ping -c 4 8.8.8.8  
ping -c 4 example.com
```

- IP works but hostname fails → investigate DNS.
- A failed ping does NOT prove the server is down; ICMP may be blocked.

DevOps mindset: use ping as one signal, not as the final diagnosis.

SSH to EC2



```
chmod 400 mykey.pem
```

```
# Amazon Linux 2023
```

```
ssh -i mykey.pem ec2-user@<public-ip>
```

```
# Ubuntu
```

```
ssh -i mykey.pem ubuntu@<public-ip>
```

- Default username depends on the AMI.
- Use the private key associated with the instance.
- The instance must be reachable and SSH traffic must be allowed (normally TCP/22).
- Permission denied? Check username, key, and key-file permissions.

Do not open SSH to 0.0.0.0/0 unless there is a justified temporary need. Prefer a narrow trusted source.

wget: Download Files



wget is designed primarily for non-interactive downloading from URLs.

```
wget https://example.com/file.zip  
wget -O artifact.zip https://example.com/download
```

- `-O` chooses the local output filename.`
- Useful for scripts that fetch artifacts or static files.

Avoid hard-coding old tool versions in a fundamentals lab; use current vendor install guidance when installing real tools.

curl: HTTP and Data Transfer



```
curl https://example.com
curl -I https://example.com
curl -L https://example.com
curl -o page.html https://example.com
curl -fL -o artifact.zip <URL>
```

- `-I` headers only`
- `-L` follow redirects`
- `-o` choose output filename`
- `-f` fail on HTTP errors`
- `-sS` quiet output but show errors`

curl becomes especially valuable later for APIs, health checks, installers, CI/CD and Kubernetes.



Archives & Getting Help



tar: Archive and Compress



tar primarily creates/extracts archives. Compression can be added, for example with gzip using -z.

```
tar -cvf backup.tar app/  
tar -xvf backup.tar
```

```
tar -czvf backup.tar.gz app/  
tar -xzvf backup.tar.gz
```

- `c` create archive
- `x` extract archive
- `v` verbose
- `f` archive filename
- `z` gzip compression

A .tar file is an archive; .tar.gz is a tar archive compressed with gzip.

unzip + AWS CLI v2 Example



Use a real AWS artifact to practice curl and unzip together.

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"  
unzip awscliv2.zip  
sudo ./aws/install  
aws --version
```

AWS also publishes an ARM64 installer. Choose the package that matches the machine architecture.

```
uname -m
```

wget or curl?



Both can download files. Their strengths differ.

wget

- Simple file downloads
- Recursive/mirroring features
- Convenient for non-interactive retrieval

curl

- HTTP headers/status
- APIs and request methods
- Uploads and many protocol workflows

DevOps rule: learn both. Use the tool that makes the task clearer and easier to automate.

Getting Help: --help and man



```
ls --help  
cp --help  
man ls  
man ssh
```

- `--help` usually gives a quick usage summary and available options.
- `man <command>` opens the manual page when installed.
- Inside man: `/word` searches, `n` finds the next match, `q` quits.

Use documentation before guessing command flags.

Finding Commands



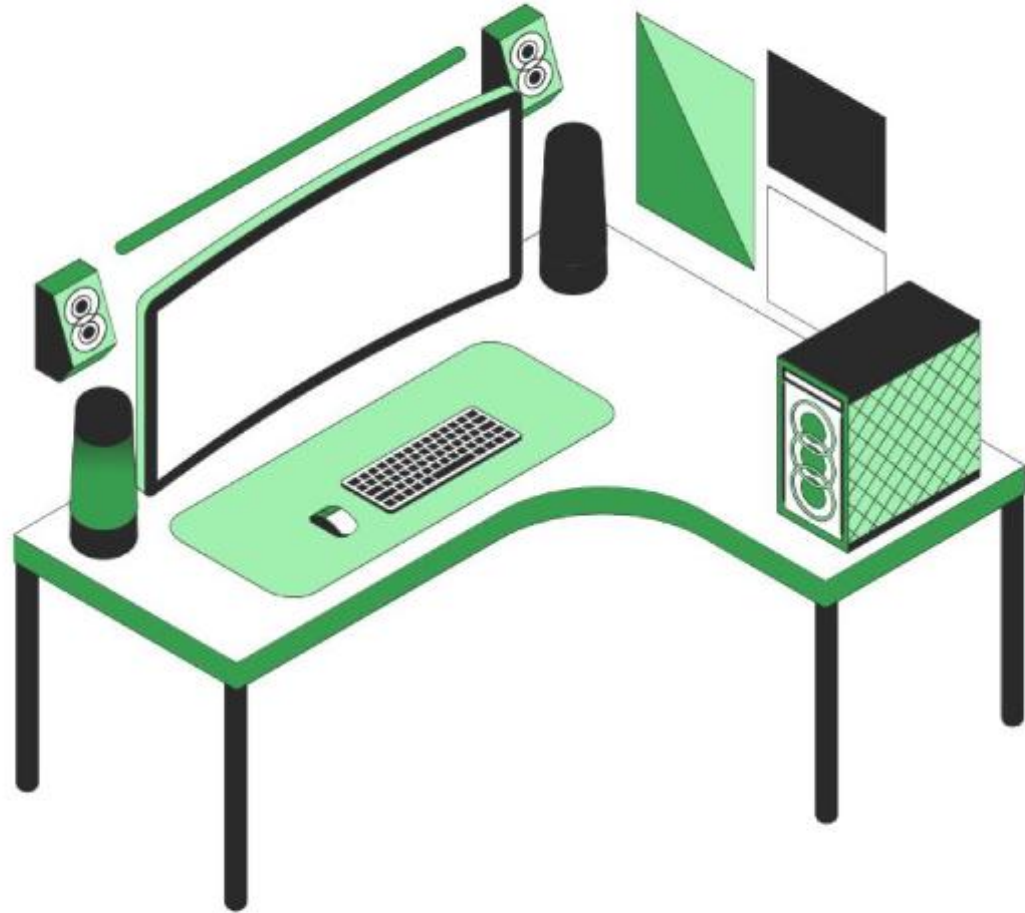
```
info coreutils  
type cd  
type ls  
which ssh  
command -v curl
```

- ``info`` provides GNU documentation where available.
- ``type`` tells how the shell resolves a command (builtin, alias, function, executable).
- ``which`` commonly prints the executable found in PATH.
- ``command -v`` is a portable shell way to locate/identify a command.

Example: ``cd`` is normally a shell builtin, so ``type cd`` is more informative than ``which cd``.

Day 2 Challenge

- Create an app directory and make a backup copy of its config.
- Test reachability to a hostname and explain what a failed ping does—and does not—prove.
- Write the correct SSH command for Amazon Linux 2023 and Ubuntu.
- Download the AWS CLI ZIP with curl, extract it, and verify the installation.
- Find help for ssh without searching the web.



Do you have any questions?

Send it to us! We hope you learned something new.